

# Pratheek Hadse Pandesha

Ph.D. Candidate & William H. Danforth Plant Sciences Fellow,  
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## EDUCATION

- 2021- 2026 **Washington University in St. Louis**  
(expected) Ph.D. Candidate, **Donald Danforth Plant Science Center** |  
Advisor: [R. Keith Slotkin](#) | Thesis: Primary small RNAs that initiate epigenetic silencing in Arabidopsis
- 2018-2020 **Indian Agricultural Research Institute, New Delhi**  
M.Sc., in Plant Physiology | CGPA – 8.65 | [Gold Medal](#)  
Advisor: [Viswanathan Chinnusamy](#) | Thesis: Investigating the role of DUF1645 proteins in abiotic stress tolerance and grain yield in rice
- 2014-2018 **University of Agricultural Sciences, Bangalore**  
B.Sc., Agriculture Biotechnology | CGPA-8.66  
Advisor: [R. Sowdhamini](#) | Thesis title: Structural and functional annotation of multiple stress responsive hypothetical proteins in rice

## RESEARCH AND PROFESSIONAL EXPERIENCE

- 2021-2022 **Washington University in St. Louis / Donald Danforth Plant Science Center** (*Graduate research rotations*)
- Targeted methylation using CRISPR / SunTag system for viral disease resistance in Arabidopsis | Advisor: [Rebecca Bart](#)
  - Effect of nitrogen on circadian regulation of lipid metabolism in Arabidopsis | Advisor: [Dmitri Nusinow](#)
  - RNA-guided DNA insertion with CRISPR-associated transposases | Advisor: [R. Keith Slotkin](#)
- 2020-2021 **Bayer Crop Science**  
*Competitive Intelligence Analyst*, Science Competitive Intelligence & Prospecting
- 2019-2020 **Massachusetts Institute of Technology (MIT), Cambridge, USA**  
*Khorana Scholar* at the Dept. of Biology & Whitehead Institute for Biomedical Research  
Advisor: [Mary Gehring](#) | Title: Pathways regulating genome dosage sensitivity in the endosperm of Arabidopsis
- 2017-2018 **National Centre for Biological Sciences (NCBS), Bangalore**  
*Bachelor's Dissertation*  
Advisor: [R. Sowdhamini](#) | Thesis title: Structural and functional annotation of multiple stress responsive hypothetical proteins in rice
- 2017 **Centre for Cellular and Molecular Biology (CCMB), Hyderabad, India**  
*JNCASR- Summer Research Fellow* | Advisor: [Imran Siddiqi](#)  
Development of fluorescent tag-based cell cycle marker system in Arabidopsis

- 2016 **University of Delhi, (South Campus), New Delhi, India**  
*Indian Academy of Sciences Summer Research Fellow* | Advisor: Surekha Agarwal  
 Identified and cloned a tetraspanin gene from *Oryza sativa*.
- 2016 **University of Agricultural Sciences, Bangalore, India**  
*Summer student* | Advisor: Veena S. Anil, Dept. of Plant Biotechnology  
 Standardized media composition and tissue culture protocol for somatic embryogenesis of Potato (Var. Kufri Jyothi- Late blight resistant cultivar)

## PEER REVIEWED PUBLICATIONS

- Montgomery, J. S., Soni, N., Marques Hill, S., Morran, S., Patterson, E. L., Edwards, S. A., ... **Pandesha, P.H.**, Slotkin, R.K. and Napier, R., & Gaines, T. (2025). A transposable element insertion in AUX/IAA16 disrupts splicing and causes auxin resistance in *Bassia scoparia*. *The Plant Journal*, 123(2), e70339.
- Liu, P., Panda, K., Edwards, S.A., Swanson, R., Yi, H., **Pandesha, P.**, Hung, Y.H., Klaas, G., Ye, X., Collins, M.V. and Renken, K.N., 2024. Transposase-assisted target-site integration for efficient plant genome engineering. *Nature*, 631(8021), pp.593-600.
- Adavi B S, **Pandesha PH**, B J, Jha SK, Chinnusamy V, Sathee L. Nitrate supply regulates tissue calcium abundance and transcript level of Calcineurin B-like (CBL) gene family in wheat. *Plant Physiol Biochem*. 2023 Jun; 199:107724. doi: 10.1016/j.plaphy.2023.107724.
- Vineeth TV, Krishna GK, **Pandesha PH**, Sathee L, Thomas S, James D, Ravikiran KT, Taria S, John C, Vinaykumar NM, Lokeshkumar BM. Photosynthetic machinery under salinity stress: Trepidations and adaptive mechanisms. *Photosynthetica*. 2023 Apr 13;61(1):73-93.
- Sathee L, Jagadhesan B, **Pandesha PH**, Barman D, Adavi B S, Nagar S, Krishna GK, Tripathi S, Jha SK, Chinnusamy V. Genome Editing Targets for Improving Nutrient Use Efficiency and Nutrient Stress Adaptation. *Front Genet*. 2022 Jun 14; 13:900897.
- Pandesha PH**, Nagar S, Advani S, Chinnusamy V, Pandey R. Functional annotation of hypothetical proteins involved in multiple stress response in *Orzya sativa*. 2021; *Ind J Agr Sci* 91, 7.

## MANUSCRIPTS UNDER REVIEW

- Pandesha, P.H.**, Sajeevan, S., Sandeep, A., Mohamed. S. K., Pandey, R., Sathee, L., Ramanathan, Sowdhamini., Chinnusamy, Viswanathan. DUF1645 is a novel intrinsically disordered transcription factor family specific to plants. (*under review*)

## CONFERENCE PRESENTATIONS

- Pandesha, P.H.**, Edwards, S.A., Ratnayake, S., ... Slotkin RK. Primary sRNAs that initiate RNAi and *de novo* RdDM, *Keystone Plant Epigenetics and Epigenome engineering meeting 2025, Fort Collins*; 13-16 Oct 2025; (*Poster*)
- Pandesha, P.H.**, Edwards, S.A., Ratnayake, S., Slotkin RK. Quest for primary sRNAs that initiate epigenetic silencing in plants, *ICAR 2024, San Diego*; 15-19 July 2024; (*Poster*)
- Pandesh P.H**, Sandip Adavi, ...Sowdhamini R, Viswanathan Chinnusamy. DUF1645- a novel transcription factor family in plants. *International Plant Physiology Congress (IPPC)*, Lucknow, India; 2-5th December 2018; (*Poster*)

## FELLOWSHIPS AND GRANTS

|             |                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 - 2026 | William H. Danforth Plant Sciences Fellowship, Washington University in St. Louis                                                                      |
| 2021        | Trinity Henry-Barlow Fellowship and Cambridge Trust Scholarship for doctoral studies at the University of Cambridge, Cambridge Trust, UK (Not availed) |
| 2020 - 2021 | ASPB Conviron Scholar, American Society of Plant Biologists                                                                                            |
| 2018 - 2020 | Junior Research Fellowship, Indian Council of Agricultural Research (ICAR)                                                                             |
| 2019        | Khorana Scholar at MIT, funded by IUSSTF, United States Government and Department of Biotechnology, Government of India                                |
| 2018        | DBT- JNU Fellowship (for master's), Department of Biotechnology, Govt. of India (Not availed)                                                          |
| 2014 - 2018 | Merit Scholarship, University of Agricultural Sciences, Bangalore                                                                                      |
| 2014 - 2018 | Fee Concession and Merit Scholarship, Government of Karnataka, India                                                                                   |
| 2014 - 2018 | SDM Education Trust Scholarship, SDM Education Trust                                                                                                   |
| 2017        | JNCASR- Summer Research Fellowship, Jawaharlal Nehru Center for Advanced Scientific Research                                                           |
| 2016        | Indian Academy of Sciences Summer Research Fellowship, Indian Academy of Sciences                                                                      |

## AWARDS AND HONORS

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|------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025 | Best oral presentation award- Donald Danforth Plant Science Center retreat                                                                         |
| 2024 | Best oral presentation award- WashU-Plant and Microbial Biosciences retreat                                                                        |
| 2024 | Danforth Center CSTM professional development award                                                                                                |
| 2024 | The International Conference on Arabidopsis Research, San Diego- Registration Fee Award, The North American Arabidopsis Steering Committee (NAASC) |
| 2021 | Gold medal and best student of the year, Indian Agricultural Research Institute, New Delhi                                                         |
| 2020 | ASPB Plant Biology 2020 (Virtual) Registration Fee Award, American Society of Plant Biology                                                        |
| 2019 | Best poster award - National Agricultural Science Congress, Indian Agricultural Research Institute, New Delhi                                      |
| 2017 | Travel grant- Nobel Prize Series, India by The Nobel Foundation, Sweden                                                                            |
| 2017 | Travel grant- Indo-German Workshop on Plant-Insect Interactions Across Gradients, NCBS, Bangalore                                                  |
| 2017 | Travel grant-Advanced Biology Lecture Series (Theme: Gut Microbiome), NCBS, Bangalore                                                              |
| 2016 | Travel grant-Advanced Biology Lecture Series (Theme: Cancer Biology), NCBS, Bangalore                                                              |
| 2014 | Inspire Internship, Department of Science and Technology, Government of India                                                                      |

## PROFESSIONAL ASSOCIATIONS AND SERVICES

|                |                                                                                                                                                                                                                                                                              |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025- Present  | WashU-DBBS Peer mentor                                                                                                                                                                                                                                                       |
| 2025           | Panelist on a career panel for REU summer students at the Danforth Center                                                                                                                                                                                                    |
| 2025           | Rotation student mentor, Slotkin Lab (Summer 2025)                                                                                                                                                                                                                           |
| 2024 - 2025    | Member, DBBS Student Advisory Committee, Washington University in St. Louis                                                                                                                                                                                                  |
| 2024 - Present | Member, North American Arabidopsis Steering Committee (NAASC)                                                                                                                                                                                                                |
| 2024 - Present | Arts & Sciences Graduate school cross program peer mentor, Washington University                                                                                                                                                                                             |
| 2023 - 2024    | Co-chair, DBBS Student Advisory Committee, Washington University in St. Louis                                                                                                                                                                                                |
| 2023 - Present | Member, National Center for Faculty Development & Diversity (through WashU-DBBS)                                                                                                                                                                                             |
| 2022 - 2023    | DBBS peer mentor, Washington University in St. Louis                                                                                                                                                                                                                         |
| 2022           | Graduate Teaching Assistant, BIO112-Introduction to Problem-Based Learning in Biology, Washington University in St. Louis                                                                                                                                                    |
| 2020 - Present | Member, American Society of Plant Biologists                                                                                                                                                                                                                                 |
| 2020 - 2023    | Scholar Mentor, WINStep Forward                                                                                                                                                                                                                                              |
| 2020           | Volunteer, GCCR-Global Survey on Respiratory Illnesses and Changes in Smell, Taste, and Flavor in response to COVID-19, Bangalore                                                                                                                                            |
| 2018 - Present | <i>Co-founder, Next Generation Scientist Foundation (NGSF)</i> , India; Working towards providing internship opportunities at world-class labs for talented undergraduates from challenging social and economic backgrounds ( <a href="http://www.ngsf.in">www.ngsf.in</a> ) |
| 2017           | <i>Rural Agricultural Work Experience Program, Karnataka, India</i><br>Work focused on analyzing the effect of late blight of potato and soft rot of ginger on shifting cropping patterns in rural parts of Karnataka state in India.                                        |
| 2016 - 2017    | Student ambassador, iBiology, Bangalore                                                                                                                                                                                                                                      |